



Enhancing Student Academic Performance with Analytics Dashboards

How data, Python, and Power BI combine to reveal insights that improve learning outcomes and enable proactive support.



Technology Driving Academic Success



Python for data work

Powerful libraries (Pandas, NumPy, SciPy) for cleaning, aggregating, and modeling student data.



Power BI for visualization

Creates interactive reports, filters, and KPIs that educators can explore without coding.



Integrated ecosystem

APIs and ETL pipelines connect LMS, attendance, and assessment systems for timely insights.



Objective: Why Analyze Student Performance?

Early risk detection

Spot students who need intervention before grades decline.

Trend identification

Reveal course- and cohort-level patterns to inform teaching strategies.

Student empowerment

Provide learners with personal progress views and actionable next steps.

Data-driven policy

Guide administrators on resource allocation and curriculum adjustments.

Scope of the Analytics Dashboard

Comprehensive view across data sources and levels of aggregation to support individual coaching and institutional planning.



Data Sources

Grades, attendance, LMS activity, assessment metadata, and demographic info.



Tracking

Individual timelines, cohort trends, and semester comparisons.



Visualizations

Score distributions, heatmaps of engagement, trend lines, and KPI tiles.



Access

Role-based views for students, tutors, instructors, and admin staff.

Methodology Overview

1. Data collection Extract logs and records from LMS, SIS, attendance systems, and assessments.

2. Preprocessing Clean, normalize, and join datasets using Python (Pandas), handle missing values and timestamps.

3. Analysis Compute descriptive stats, percentiles, and segment by demographics and course.

4. Reporting Publish interactive Power BI dashboards with scheduled refresh and role-based access.



Python in Action: Data Processing & Analysis

Python automates preparation, analysis, and feature engineering: aggregating scores, computing moving averages, detecting outliers, and producing cohort summaries for visualization.

Data cleaning

Normalize grade scales, impute missing attendance, timestamp alignment.

Feature engineering

Create engagement scores, rolling GPA, course difficulty indices.

Automated pipelines

Scheduled scripts push curated datasets to Power BI for near real-time reports.



Power BI Dashboard Features



- Interactive filters: course, semester, demographic, instructor.
- Visuals: grade distributions, attendance heatmaps, engagement scatter plots.
- Alerts & KPIs: automated flags for at-risk students and threshold breaches.
- Export & sharing: snapshots for advising, secure publish to faculty groups.



Impact & Benefits

Retention & success

Early intervention increases retention and improves completion rates.

Personalized support

Tailored tutoring and resources based on individual performance signals.

Transparency & motivation

Students see progress and action steps, boosting accountability.

Institutional insight

Data guides curriculum improvement and strategic allocation of resources.



Conclusion: The Future of Academic Analytics

Move from descriptive to predictive analytics4integrate AI models to forecast risk, include wellbeing and social engagement signals, and continually refine dashboards through educator feedback. Call to action: adopt data-driven practices to unlock each student9s potential.

❏ Next steps: pilot a dashboard for one department, validate predictive models, and scale with automated data pipelines.